



Nodes of certainty and spaces for doubt in AI ethics for engineers

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ABSTRACT

Discussions about AI development frequently bring up the question of ethics because it is difficult to predict how technological decisions might play out once AI systems are implemented and used in the world. Engineers of AI systems are increasingly expected to go beyond the traditions of requirement specifications, taking into account broader societal contexts and their complexities. In this paper we present findings from a hackathon event conducted with working engineers, exploring the gaps between existing guidelines and recommendations for addressing ethical issues with respect to AI technologies and the realities experienced by the engineers in practice. We found that when faced with the uncertainties of how to recognize and navigate ethical issues and challenges, engineers looked to identify the responsibilities that need to be in place to sustain trust and to hold the relevant parties to account for their misdeeds. We re-envision familiar engineering practices as nodes of certainty to accommodate the needs of responsible and ethical AI. Despite the desire for mechanisms for sustaining certainty in how to build AI technology responsibly by providing frameworks for action and accountability, there remains a need to ensure just enough spaces and opportunities to cultivate reasonable doubt. Space and capacity to doubt accepted certainties, in fact, is the very process of ethics, necessary for holding to account our standards, guidelines, and checklists as technology and society co-evolve.

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Introduction

The rapid evolution of a broad range of intelligent and autonomous systems (AI)¹ holds much potential for economic growth and vast social benefit, but also well-founded concerns about the ethical, legal and social risks these technologies represent. The vigorous debates about the importance of ethical considerations in the design and development of AI have resulted in a multitude of frameworks, guidelines, principles, and regulations intended to provide a foundation for better technologies of the future (Jobin et al., 2019). Yet there remain important gaps between these efforts for addressing ethical issues in the development of AI systems and the realities of engineering practice.

In this paper we present findings from a hackathon² event conducted with working engineers, exploring the gaps between existing guidelines and recommendations for addressing ethical issues with respect to AI technologies and engineering practice. We found that when faced with the uncertainties of how to recognize and navigate ethical issues and challenges, engineers looked to identify the responsibilities that need to be in place to sustain trust and to hold the relevant parties to account. Familiar engineering practices can be augmented and re-envisioned as *Nodes of Certainty* to accommodate the efforts towards responsible and ethical AI. Despite the desire for mechanisms for sustaining certainty in how to build AI technology responsibly, there remains a need to ensure enough spaces and opportunities to cultivate reasonable doubt. We argue that the space and capacity to doubt accepted certainties is the very process of ethics, necessary for holding to account our standards, guidelines and checklists as technology and society co-evolve.

We advocate for a participatory process that is essential to ethics. Such a process requires collective action and openness, necessitating constant engagement with the world. To explore these aspects of ethics, we turn to Bakhtin's (ca. 1920, 1993) emphasis on (1) the value of doubt in ethics as a form of participation, and (2) the *unfinalizability* of ethics that characterizes its open-endedness. Complex and multi-faceted problems such as ethical decision-making in AI development necessarily require an iterative process to develop infrastructures of certainty. We refer to Inger Sjørlev's (2017) notion of *ritual infrastructures* to capture how countering doubts and uncertainties is an iterative, ritualistic process enabling these infrastructures. This process is constantly transformed to align with the changing circumstances and contexts of social life, becoming a mechanism of managing the tensions between certainty and doubt. In AI development, such infrastructures are what we term nodes of certainty that AI practitioners seek when dealing with ethical dilemmas.

In what follows we review extant research on ethical AI standards and certifications, providing a general backdrop of the current status of ethics in AI development. Drawing on Bakhtin's philosophy of the act and anthropological literature on faith and ritual, we theorize spaces for discussion and doubt as key to enabling ethics in action and in engineering practice. Debates situated within spaces for doubt can help establish trust and certainty through the ritualization of the nodes of certainty. The iterative process of movement between nodes of certainty and spaces for ethical discussion provides an initial framework for how approaches to AI ethics may be implemented in engineering practice.

AI ethics – the state of the debate

There is a growing scholarly literature comparing AI ethics guidelines, principles, and standards produced by companies, civil society organizations, and government expert groups (Jobin et al., 2019; Mahieu et al., 2018; Whittlestone et al., 2019). Although most documents focus on a similar set of core concepts, such as accountability, transparency, and privacy among others, there is great divergence in what is emphasized and in which context (Jobin et al., 2019). This comparative research reveals how emerging AI ethics guidelines and documents have changed over time (Greene et al., 2019; Mittelstadt et al., 2016; Schiff et al., 2020). While the process of creating these documents is an

important site of ethics in action, the documents themselves are necessarily simplifications that narrow down what counts as important considerations in AI development (Green & Viljoen, 2020). Despite the inherent simplifications, the AI ethics guidelines, principles, and standards remain difficult to translate into practice (Metcalf et al., 2019; Mittelstadt, 2019). As organizations such as ISO and IEEE embark on standards development, researchers warn that the appearance of consensus around shared ethical principles may in fact hide political and normative disagreements, thus failing to guarantee development of ethical and trustworthy AI systems (Green & Viljoen, 2020; Mittelstadt, 2019).

The struggle to bridge the gap between theory and practice is not unique to the domain of AI ethics, where current attempts involve calls for development of robust approaches to governance, stronger regulation, and clearer standard setting. While these are important efforts, they can never be complete, nor should they be. Ethical decision making is fraught with uncertainty and doubt because of the inevitable value tensions generated when abstract principles meet practical challenges. Practitioners must make trade-offs and prioritize some values over others when accounting for current technological capabilities and constraints (Whittlestone et al., 2019). These tensions are reminiscent of what sociologist Christopher Groves terms ‘the double bind within the “ethical life”’ (2015, p. 322), in which we are reflexive about what we do, but our practices are also formed by the knowledge that makes our actions possible in the first place, constrained by the uncertainties of predicting the future.

Even where particular principles or guidelines are followed, there remains a need for space and capacity to locate who and what to hold accountable if unwanted outcomes arise (Wieringa, 2020), or accounting for complexities of applying these guiding statements to various contexts (Beaudouin et al., 2020; Whittlestone et al., 2019). It seems the work of ethics is located in the navigating between abstract principles and concrete practices. The inherent uncertainty of such a process is rooted in the limits of what is currently possible and what is available (Groves, 2015). Open-endedness is an important part of the tension that arises when dealing with ethical considerations, allowing for adjustments to frameworks while, at the same time, acknowledging that developments in ethical governance need to remain open to change (Kroll et al., 2016; Wieringa, 2020). Perhaps the ethical process needs to be recognized as one working through gaps, rather than resolving them.

Engineering communities seek certainty in ethical decision-making with respect to AI development, often turning to standards and checklists or seeking counsel from entities that are in possession of relevant expertise (Metcalf et al., 2019) as expert adjudicators, taking on the responsibility for the questions of right and wrong. The typical risk-based analysis approaches, however, are often ill equipped to deal with uncertainties stemming from inevitable value tensions inherent in the development of technical systems (Groves, 2015). As ethical considerations enter organizations and become an institutionalized set of roles and responsibilities, this can distance the work of ethics from the practice of technology development (Metcalf et al., 2019). To counter this, several recent studies have explored engaging practitioners in co-creating ethical AI guidelines and checklists. Madaio et al. (2020), for instance, found that practitioners participating in checklist co-design were able to identify tensions and engage in generative critical conversations. Pillai et al. (2021) suggest co-designing ethics already in higher education to

urge future collective participation in the work of ethics. Such direct engagements with practitioners are important, but it is clear that shifting the responsibility for ethical reasoning solely on individual engineers is an untenable proposal (ANE, 2018). Debates about ethical decision-making must happen at different levels within organizations and not rest exclusively with the developers themselves.

Groves (2015) argues that current technical innovation does not require better rules, but calls for changes to practices, to the institutions that sustain them, and to the social relationships built around them. While a sense of certainty is necessary when making decisions in the development of technology, such certainty must remain open to challenge as both technology and society are changed through innovation.

Embracing the unfinalizability of ethics

Rather than providing definitive answers to what is a right or a wrong decision, ethical frameworks offer ways of thinking and reasoning through doubts about which decision is better given the context (Frauenberger et al., 2017; Ustek-Spilda et al., 2019). Here we rely on Bakhtin (ca. 1920, 1993) to interpret the tension between ethical principles and practice as one that is characterized by seeking certainty through the struggles with uncertainty and doubt. We conceptualize the practice of ethics as the difficult and generative process of deliberation and debate that characterizes the movement from doubt to certainty in decision making. Bakhtin highlights the *unfinalizability* of ethics and the constant engagement with uncertainty and doubt as a route to the development of a kind of moral wisdom (Morson & Emerson, 1990).

In his essay *Toward a Philosophy of the Act* Bakhtin considers the abstract laws of behaviour and the generalized theoretical judgements for how we *ought* to act and finds them unhelpful. He argues that moral behaviour is not a matter of following rules, challenging the Kantian notion of duty in his writings. To Bakhtin, the individual act is far more than merely an instantiation of a generalized normative directive (or a failure to do so). Rather, the act itself is a complex expression of situated moral wisdom, more reminiscent of Aristotelian virtue ethics (Ustek-Spilda et al., 2019; Vallor, 2011). Bakhtin argued that an over-emphasis on logic and structure overlooks the diversity and particularity essential to understanding life. This informed his position on ethics, where in his view ‘all approaches to ethics in terms of rules not only ignore essential particulars that fail to fit a rule, but also function in a fundamentally mechanical way’ (Bakhtin, 1993; Morson & Emerson, 1990, p. 25).

The uncertainty and doubt inherent in ethical reasoning are the result of attempting to act responsibly in a world that is always far more idiosyncratic, open and messy than what is envisioned through rule-based logic. To Bakhtin, if ethics were merely a matter of following rules, then all we need is to follow the relevant rules and all the work of ethics would disappear in their thoughtless application. Yet rules, with all their inflexibility, must always be adapted and interpreted situationally. Ethical principles, rules, standards, and guidelines are important in that they can lay out possibilities of action and its constraints. Ethics in practice, however, is always located in particular situations where the real work of judgement must be done. To Bakhtin, this work of judgement is the essence of an ethical act. It is necessarily risky and must involve engagement with the particular situations and people with their own idiosyncrasies (Morson & Emerson, 1990).

Bakhtin's (1993) approach to moral action is rooted in dialogics – the act of speaking as a moral, participative action that reveals itself in relation to a predominant notion of morality. This is what Bakhtin refers to as *answerability*, an individual responsibility to act in the world with and in response to others. Here, doubt constitutes a distinctive value that points to the importance of the particular in decision-making and judgement calls. Ethical issues are imbued with doubt. The process of searching for solutions is inherently a process intended to counter the doubt intrinsic to ethical dilemmas, but it must resist the false idea of a one-size-fits-all solution, which is unattainable. Following the Bakhtinian approach to speaking as an act that is participatory, we understand ethics in action as a collective process that is enabled by individual participation in moral reasoning through debate and discussion.

Bakhtin uses the idea of *unfinalizability*³ to emphasize that the world is an open and messy place, which can never be fully described and controlled through formal structures and logics (Bakhtin, ca.1920; Morson & Emerson, 1990). There is freedom in this openness, but it cannot be without constraint. Ethics too is a balance between openness and constraint – unfinalizable but striving towards some level of finalizability. Though for Bakhtin ethics remain the responsibility of the individual, it is not the purview of individual reasoning alone – the striving for finalizability requires engagement with and accounting for another as our moral sense becomes enriched through experience and dialogue (Bakhtin, ca.1920; Morson & Emerson, 1990). The balancing between rule-following and open-ended deliberation is key to sustainable approaches to ethics in AI. Ethical deliberation requires conscious intention, effort, and communal engagement. Seen from this vantage, the vigorous debates surrounding AI ethics are, in and of themselves, manifestations of necessary ethics in action. This striving for finalizability through checklists and standards must be balanced with an understanding that where ethics is concerned, a definitive framework will remain forever unreachable and rightly so.

Enabling possibilities: multiple paths to certainty

In an engineering world of risk minimization and performance optimisations, the ability to be provably certain about potential outcomes is the ultimate goal. It is no wonder then that engineers often clamour for certainty achieved through concrete guidelines and well-specified rules. Where dialogic ethical deliberation is full of uncertainty, requiring significant overhead, checklists, and standards leave little wiggle room and can be quickly completed (Patil et al., 2018). The problem with institutionalizing ethics, whether in open-ended questions or closed checklists, lies in severing a personal engagement with ethics. Instead of approaching ethics as a preventive measure to guard against potential future consequences, an ethics that offers structured and safe spaces for deliberative discussion can offer more possibilities for people to participate and to develop their moral sense over time (Pink, 2019). This approach offers opportunities for those involved to speak out and challenge existing governance structures, addressing their potential limitations and incommensurability with the particulars of this or that situation. What is necessary is a particular kind of participation that allows individuals to voice their doubts when in search of certainty.

Anthropologist Inger Sjørlev refers to *ritual infrastructures* as a means of attaining certainty amid doubt, as 'structures that form subjectivity and sustain efforts to achieve

some kind of existential certainty' (2017, p. 66). She suggests that rituals provide paths to certainty that are undergirded by the spatial location, the material requirements, and the character of ritual practice. Infrastructures of certainty, in the case of Sjørølev's informants, were sustained by externalizing peoples' subjective relationships to their beliefs through material, spatial, and institutional means. The iterative process of negotiating doubts became integral to attaining or reaffirming a sense of certainty – doubting is the built in and expected part of the process. If we accept that doubt is an unavoidable part of ethical practice and cannot be eradicated, embracing doubt may be one way to cultivate certainty. Certainty and doubt are processes that co-construct one another as two sides of the same ethical coin. As Bakhtin would have it, if doubt is to be embraced in its own right, there needs to be room to engage with doubt through participation. Through providing communal spaces for negotiating doubt, we can support different paths to certainty in AI development. The debates and discussions at the AI hackathon we organized evinced a need for infrastructures for dealing with ethical doubts and dilemmas by establishing a sense of certainty that can enable action.

'Ethical dilemmas in AI' Hackathon

In September 2020, the authors collaborated with two engineering trade organizations, the Association of Nordic Engineers (ANE) and the IEEE, in addition to the DataEthics ThinkDoTank. We organized an online hackathon 'Ethical Dilemmas in AI' for practicing engineers to discuss ethical considerations in AI development and what challenges these present in engineering practice. The event attracted an international group of 18 engineers from industry and academia, actively participating in online activities and workshops spread out over two days. The event was recorded for use as ethnographic material with the agreement of the participants. However, due to data protection requirements, the authors never had access to the registration details of the hackathon participants beyond minimal information.⁴ All of the group work and discussion sessions were recorded and the authors took extensive notes throughout. The video-recordings were transcribed by the second author ensuring anonymization. After transcription all of the recordings were erased to ensure confidentiality. The authors also conducted four post-hoc interviews with participants who volunteered, exploring their experience of the hackathon as well as their opinions and concerns regarding the opportunities and challenges inherent in AI development. The interviews were video-recorded and transcribed by the authors. Our analytical approach relied on an iterative analysis of the transcripts through open and focused thematic coding conducted by the second author with weekly discussions of progress and insights.

Although hackathon participants agreed on the importance of ethical considerations, they struggled with how to identify what constitutes an ethical issue in practice and, if identified, how to address such. There is a clear need for engineers working with AI to have organized spaces for discussion around ethical dilemmas as a way to work through mounting uncertainties. The lack of opportunities for discussion was a clear marker that little was being done to pinpoint the particular practices of AI practitioners and to locate who or what should be held accountable if unwanted outcomes were to emerge and how this ought to be accomplished. AI practitioners currently face many uncertainties when having to deal with ethical challenges. We argue that it is imperative to establish

mechanisms and infrastructures for certainty through (1) *spaces for ethical debate*, and (2) *nodes of certainty*. Spaces for ethics allow engineers to discuss their dilemmas. In doing so they also collectively build a system of support through which to pave paths towards particular nodes of certainty, which are tangible steps in AI development for putting ethics into practice while considering contexts and current possibilities.

During the hackathon it became clear that many participants were not certain how, when and in what way existing guidelines might be applied, who is responsible for their application, and who might be in a position to evaluate outcomes. More importantly, they typically had few opportunities to discuss whether and how these guidelines might be useful and applicable in practice. Here we seek to embrace the tension between AI ethics governance and developer practices. This perspective acknowledges practitioner needs for having guidelines to hand when dealing with ethical dilemmas and encourages critical discussions among practitioners that can work to counter uncertainties and doubts.

Negotiating doubt and certainty in AI ethics

The goal of the 2020 AI Hackathon was to provide a space for engineers to discuss what kinds of concrete and actionable practices could help address ethical issues and ambivalences in their own work with AI systems. The hackathon participants were assigned to groups to discuss real-life cases of ethical dilemmas in particular contexts (ANE, 2021). Engineers accounted for current practices, possibilities for ethical considerations in practice, and the available resources to do so.

Time and again discussion circled back to a lack of clarity for whom to address when ethical issues are identified, dovetailing with a confusion about what ought to happen afterwards. This overwhelming concern with the perceived lack of institutionalized frameworks for ethical deliberation pushed the conversation towards debates about governance.

Participants situated practices within their immediate contexts, but also took into account the responsibilities of actors in governmental, municipal, and corporate contexts and the effect they may have on the more local-level decision-making in their workplaces. Ethical concerns carry with them a burden of many responsibilities. In our discussions many participants felt that they did not have the capacity to take on these responsibilities. Most engineers operate within hierarchies of power and our participants were uncertain about what infrastructure of support they have for preforming their responsibilities and being held accountable in terms of ethical concerns.

It would be nice to have some sort of independent body or a person or organisation to whom you could report or with whom you could discuss about these ethical concerns that you're having. (Day 2, group 2, plenum discussion).

When faced with uncertainty, participants pointed towards what they felt may provide them a sense of certainty and clarity. They sought a loose framework for which actions can be judged as ethical or unethical, thus looking to develop what Bakhtin calls a situated moral wisdom (Morson & Emerson, 1990). At the same time, they wanted concrete guidance for what to do in the face of problems. General ethical concerns, such as ensuring fairness or treating all people with dignity, are far too broad for a single solution. In

practice such concerns must be addressed one context at a time, paying attention to the idiosyncratic particulars of the situations and the people involved. Here we see the need for a version of Sjørsløv's (2017) ritual infrastructures for ethical deliberation and ethical practice that can enable paths to certainty. These ethics infrastructures must enable and support spaces for discussions and expressions of concern as well as provide what we call *nodes of certainty* to help manage inherent doubts of ethical reasoning. Nodes of certainty take their departure from familiar practices in engineering such as documentation and performance oversight but acknowledge the essential *unfinalizability* of the ethical process by remaining open to challenge through spaces of discussion and debate.

Making space for doubt

During the hackathon, participants struggled with doubts arising from ethical dilemmas of AI development and the difficulty of addressing these on their own, due to the level of responsibility this would place on them. Rather than shifting that responsibility elsewhere (although this was also proposed), many sought opportunities for discussion. As one participant noted:

The engineers are encountering dilemmas or issues every day, there is a lot of responsibility on the [sic] shoulders. So, we have to provide support through discussion in organisations and between the engineers. Because these are not questions that one person can give the right answer to. No. So discussions are very important. (Day 1, group 2, group discussion)

When participants discussed specific example cases in small groups, they tried to collectively work through their doubts and sought routes to certainty. These certainties were then challenged in plenum discussions, forcing the groups to work through new doubts, seeking more nuanced positions. Consider the following conversation in a working group after a plenum discussion that touched on trustworthiness and explainability of AI:

A: So I think in terms of explainability, if you say everything has to be explainable, then that's the end of most AI. I'm not saying we shouldn't explain. I'm just saying we have to be aware that our explanations might be wrong and biased.

B: yeah, and it's irritating – it's the standard engineering question – it depends on, it's a trade-off – 100% explainable AI means no machine learning, you can't have it. It's not 100% explainable. And that's bad because we don't get the benefits. Zero explainability, right, it does something we don't know what and we have no clue how it does it. That's bad too. And it's basically a middle ground, you know, you might want it to output for example, sub results or whatever. But it's an annoying slider, you can't have a "this is the answer".

A: I think as well. Someone said something interesting in the plenary that humans can't explain – we can't explain our decisions. How can we make a system that can? you know what I mean? It's one of those questions, isn't it?

(Day 2, group 3, group preparation)

Starting out with an oft-stated maxim that trustworthiness and explainability of AI systems are highly desirable properties, participants struggled with the fact that most principles and guidelines do not specify what these terms mean or how to achieve sufficient amounts of either to qualify as 'ethical.' The plenum discussion forced each group to re-evaluate their approaches and ideas. In this way, the hackathon provided a safe space for

debating difficult questions and challenging certainties built on shaky ground, in order to co-create new certainties by externalizing doubts through discussion.

Participants recognized the value of the opportunity to have the discussions, which lead to considering what discussion spaces might need to look like within and outside their workplace environments. We identified three types of such spaces for challenging certainties and working through doubts – education and training spaces, spaces for discussion, and spaces for expressions of concern.

Education and training spaces were most familiar, where engineers could participate in interdisciplinary training and assess their technologies from perspectives outside their own field. Participants often expressed that they felt a lack of competence when talking about societal issues associated with AI development. Discussion ranged from envisioning changes in traditions of secondary engineering education to life-long learning opportunities where ethics and humanities-based points of view might become part of training and courses typically offered as career development options.

Most discussions, however, focused on the need for institutionally supported spaces for collaborative discussion and debate both within workplaces and without, through trade organizations, civil society, or even political fora. Current attempts to counter doubt through standards and guidelines can often fail because there is a lack of space and time to discuss how to use them and contextualize them in practice:

There's often not enough time and knowledge in the traditional software development to consider all the impacts on ethics, or at least it's not the number one priority when the decisions and design decisions are made. (Day 2, group 2, plenum presentation)

Spaces for discussion need to become commonplace, sanctioned as part of infrastructure of ethical deliberation internally in workplace settings as well as fostering broader discussion across other organizational configurations.

And the problem in these kind of design processes is also then if the purpose is not transparent, so you might think you are developing for application x but might be that there are some other goals as well included in the process – then we agreed, but there's currently no good places for engineers to discuss these problems. (Day 2, group 2, plenum presentation)

When normalized by routinely allocating time and space to discussions that allow participants to work through doubts and concerns in an organized rather than ad hoc manner, spaces for discussion can enable the routes from doubt to certainty that Sjørølev highlights. After all, the work of ethics and judgement has to happen somewhere.

Where spaces for discussion were intended to help figure out how to do things, spaces for expressions of concern were proposed as practitioners considered the implications of ethical deliberation in a workplace context. Ethical issues are frequently sensitive, thus spaces for concern are necessary to allow practitioners to voice their doubts outside the hierarchies of the workplace. There was an emphasis on having safe spaces where mistakes can be admitted and debated. As one participant noted, there is a need for a kind of 'no blame culture that allows for failures to be publicly discussed, so that we can learn from them.' (Day 2, group 4, plenum presentation)

The way spaces for ethics work to address doubt has much to do with the issue of responsibility in AI ethics. Engineers felt overwhelmed, calling for distributed responsibility and accountability measures. To better understand how to approach the problem

with consideration for ever-changing contexts, we identified *nodes of certainty* as a way to counter doubt in AI ethics by taking departure from familiar engineering practices.

Nodes of certainty

While spaces for ethics work to locate and engage doubts in AI ethics, *nodes of certainty* are ways to tackle these doubts in concrete terms. The hackathon participants referred to their own practices within the workplace to suggest various approaches to ethical dilemmas. Overall, we identified six nodes of certainty – documentation, testing, standards, certification, punitive measures, and oversight. The first three – documentation, testing, and standards – are specific to engineering practice, and can be directly mapped to individual responsibilities. Certification and punitive measures, in turn, are forms of governance that are often imposed from outside the organization. Finally, participants discussed the issue of oversight extensively, resulting in three different types of oversight – administrative, automated, and human-in-the-loop – referring to different structures of accountability that might be deployed for these purposes. Together with spaces for discussion and concern, these nodes of certainty form an infrastructure for integrating AI ethics considerations into AI system development in practice. [Table 1](#) details the six nodes of certainty, the requirements their activation would impose, whether the responsibility for their instantiation lies internally within the workplace or externally with other types of institutions and pointing to the types of openness they require for engagement with spaces for doubt.

Documentation

Code documentation reveals the work process, but often varies in quality and levels of completeness. There was a general consensus that any effort at oversight of AI systems was largely dependent on engineering documentation. Participants agreed that documentation is an individual responsibility although it is a time-consuming task and requires relevant expectations and infrastructures to be in place:

We've already mentioned documentation. You know, you make your choice and you're documented. That's the responsibility of the individual. So you have to have that – that's where the framework needs to build. These are the things you need to do an assessment around. You make a design decision, you document it – organizationally, you need that framework that defines what those things are – the risk criteria, and then some system to monitor and maintain. (Day 2, group 3, group preparation)

While engineers are willing to take on the responsibility of documentation for the purposes of AI ethics governance, what ought to be documented, how and for which audience, remains undefined. As a task that is deeply integrated into the everyday engineering practice, documentation can be a source of certainty and an opportunity to engage ethical consideration as part of AI system development. Documentation is a ritualized set of activities that constitutes a way for engineers to externalize their thinking in the process of system development. As such, this is an opportunity for reflection and a way to make objective and concrete subjective feelings and concerns. Documentation is a form of externalization that can offer a basis for debate and dialogue as a route for collective co-creation of certainty where what is recorded and what

Table 1. Nodes of certainty in AI ethics governance.

Node of certainty	Instantiation requirements	Institutional location	Connection to spaces for doubt
Documentation	Changes in documentation practices	Internally as part of engineering practice.	Spaces for discussion: used as basis for debate. Spaces for concern: used evidence for challenge.
Testing	Restructuring of AI system development process, adding steps	Internally as part of organizational AI system development process.	Spaces for discussion: considerations for testing. Spaces for concern: testing outcomes used as evidence for challenges.
Standards	Workplace commitment for structuring AI development processes	External standards organizations. Often result of large-scale collaborative processes.	Spaces for discussion: content & assumptions carefully reviewed and open to challenge with opportunities for adjustment to particular contexts.
Certification	Individual commitment from engineers to obtain certification and workplace support	Certification programs overseen by external institutions, collectively obtained legitimacy.	Education & training spaces: provide necessary content for certification. Spaces for discussion: considerations for changes in purview given experience and needs. Spaces for concern: basis for challenge.
Oversight (administrative)	Internal changes to documentation practices to make oversight possible	Externally imposed form of governance, institutionally sanctioned, operating based on regulatory requirements.	Spaces for concern: creating a clear mechanism for expression of concern by stepping outside workplace hierarchies.
Oversight (automated)	Internal implementation of automated systems integrated into AI development practices	The technical system may be developed internally or provided as an external service.	Spaces for discussion: system assumptions, implementation and integration into practice open for debate. Spaces for concern: system output open to challenge or used as evidence.
Oversight (human-in-the-loop)	Internal changes in AI system architecture and negotiation with available humans to be in the loop	Current approaches are case-by-case negotiations. How humans should be organized for in-the-loop work is a question of justice and legitimacy.	Spaces for discussion: humans doing in-the-loop work included in debates. Spaces for concern: humans doing in-the-loop work can challenge system processes and voice other concerns.
Punitive measures	Clear articulation of responsibilities and obligations for all involved in AI system development	Spread across external organizations and internal workplace policies depending on goals.	Spaces for discussion: opportunities to challenge basic reasoning and premises of punitive action. Spaces for concern: whistle-blower protection.

is omitted can be questioned and discussed. In Sjørølev's terms (2017), such forms of externalization constitute foundations for ritualized infrastructures that enable multiple routes from doubt to certainty.

Testing

The idea of testing AI systems was voiced forcefully by a relative minority of participants. Testing technical systems prior to deployment is a common practice among

developers. The engineers discussed that testing AI systems must at least be iterated to account for emergent consequences. Others suggested that testing should involve real-time monitoring and oversight to ensure a system's adaptability throughout its lifetime:

“So when they change the algorithm, what they do is they take half a percent of the users or a very small percentage, and run the algorithm on them and check what happens right before they're implemented on the bigger states basically recognizing we cannot guarantee what will happen. So we need testing, testing, testing.” (Day 2, group 3, group preparation)

This emphasis on testing was one way to respond to the familiar contours of the Collingridge dilemma (1981), while acknowledging the difficulty of identifying potential consequences of technology at points where intervention is possible. Testing as a cornerstone of engineering processes requires internal organizational support to enable it, but it also represents a point of certainty, a source of evidence for the need to change and perhaps an opportunity for debate.

Standards

Engineering standards define and structure myriad aspects of the profession. The International Organization for Standardization (ISO) website suggests a way to think about standards as ‘a formula that describes the best way of doing something’ <https://www.iso.org/standards.html>. It is no surprise then that ISO and other technical standards bodies have been actively developing standards for trustworthy and ethical AI. For many engineers, standards represent a route to certainty and clarity, a practical guide to best practices that can leave the guesswork behind. As one participant explained:

Hopefully this kind of a discussion around “we're trying these things and that should be good” is taking place for instance, in standards development, which should lead to clear standards, and which are generally considered to be a formalisation of what is best practice. (Day 2, group 4, plenum presentation)

The hopes for someone else to find answers to difficult ethical questions is not unusual. Yet participants acknowledged that standardization is an iterative process. From Bakhtin's point of view following standards does not constitute an ethical practice but is a form of distancing from ethics through mechanical rule following. At the same time some form of guidance is necessary in a space where there are more questions than answers. If ethics is unfinalizable then the development of standards is an ethics process towards finalization that is doomed to remain incomplete. In this way, standards, while providing a route to certainty, must remain open to challenge, with a clear-eyed understanding that no standard can ever account for all the particularities and idiosyncrasies of context (Lampland & Star, 2009).

Certification

Debates about ethics in AI often bring up questions of competence. From an engineering point of view one way of dealing with changes in requirements of expertise for technology innovation is to create systems of professional licensing and certification. These become a testimony of expertise in a particular well-defined area. In concert with the idea of opportunities for life-long learning and development, the idea of certification

for AI system development in particular was voiced often as an expected obligation of engineers:

So among the other obligations for engineers is to make sure that you have the necessary skills, preferably with certification, in order to make sure that you actually produce high quality results. Such kind of certification might come from professional organisations, or from the employing organisations, especially if we're talking about larger employment bodies, such as the public sector. There is a need to establish what is the proper kind of certification and this requires public consultation, because the affected stakeholders, in this case, are the public. Such a public consultation would address what kind of certification on ethics would be necessary. (Day 2, group 4, plenum discussion)

Although there was significant debate and uncertainty about whether companies, products or particular system development processes could be labelled or certified as 'ethical' or 'ethically aligned,' the idea that individuals could be certified as 'ethically aligned' or at least in possession of required expertise, was deemed both reasonable and attainable. This led to extensive discussions about needed changes in engineering education and considerations for what working engineers might seek as learning opportunities.

Certifications are markers of expertise in a particular area. As such, conferring certifications is often done by external entities that can claim capacity to evaluate and judge appropriate levels of expertise. They are a way of 'raising the bar' as it were, for who can engage in AI system development (Weeden, 2002). They are also opportunities for changing what counts as sufficient expertise for doing things 'responsibly' and 'correctly' with respect to expectations of AI ethics. Given the essential unfinalizability of ethics, however, certification of expertise must always remain open to challenge as society and technology transform.

Oversight

One of the most common considerations in the hackathon was the question of oversight. Participants discussed the need for oversight as a form of outside intervention to help cope with the uncertainties and doubts about potential adverse outcomes. The discussion of oversight resulted from attempts to identify who must be held accountable, for what potential issues, in what way, and, most importantly, by whom.

Participants identified three different types of oversight – the traditional administrative and potentially external form of oversight, a technical solution of automated oversight that would ensure efficiency, and human-in-the-loop approaches to system development as a way to mitigate the worst problems of automated decision making. Each of these forms of oversight points to different accountabilities and forms of governance. Administrative oversight involved regulatory involvement to ensure punitive measures for unethical decisions, as well as external oversight of companies' practices. Automated oversight was to take over the monotonous tasks of risk assessments and auditing for testing and documentation. Where automating an iterative process is possible, it cannot fully take over the considerations about the future, especially if we are to maintain a level of openness about where automated decisions may lead.

Finally, the discussion of human-in-the-loop oversight circled around the desire to outsource judgement by relying on the capacity of human expertise to address the

limitations of algorithmic systems. Where human-in-the-loop is an architectural solution, it is also a means to externalize and distance ethical issues from the engineering concerns. While positive about this idea, participants pointed out its many pitfalls as well, noting that who are the humans put in the loop and what capacity they may have to influence the system or to refute it remains a question.

Punitive measures

Much of the discussion focused on consequences for unethical decisions at all levels of accountability in AI development in order to ensure responsible decision-making and compliance with standards and other directives. At the same time there was an overall agreement that a ‘no-blame culture’ should be promoted, enabling engineers to openly debate their mistakes rather than just being punished for them. One way of ensuring that engineers’ intentions aligned with ethical values was to ensure their participation in investigatory processes, as well as in collective discussions, and to monitor for evidence of criminal neglect.

The no-blame approach to punitive measures emphasizes the importance of participation as an individual responsibility, while upholding the mechanisms that enable participation is a collective responsibility. AI system development is a complex process and, as one participant put it, ‘there’s a lot of misunderstandings about how hard and weird it is to design these systems.’ (day 2, group 3, group preparation). As such, participants saw it as an engineer’s moral duty to participate in dialogue broadly within and outside their workplace to ensure people understand the technical details, limitations, and challenges of AI systems.

Conclusion

As the European Union gears up for regulating the European approach to artificial intelligence (AI HLEG, 2019), it is vital that we cultivate an understanding that ethics, whether in daily practice or technology development, will always remain *unfinalizable*, even as we strive for finalizability through debate, discussion, and creation of standards and regulations. The combination of spaces for doubt and nodes of certainty provides a framework for thinking about how any AI regulation might need to be implemented in practice from an engineering point of view. The tension between doubt and certainty is here to stay and having infrastructures in place that can support particular paths for movement between doubt and certainty is crucial.

Technology development is change-oriented, where potential risks and outcomes of innovation are never entirely predictable. The question with innovation is not whether to shoulder the risks of innovation, but, as Groves puts it, ‘how we collectively decide *which* risks and uncertainties are worth bearing’ (Groves, 2015, p. 8) and, incidentally, who ought to bear those risks. The Ethical Dilemmas in AI Hackathon aimed to identify what engineers saw as their responsibilities and obligations with respect to ethics in AI system development, seeking ideas for how these might be carried out. Ethics as a process is imbued with ambivalences around what is right and what is wrong. Echoing Bakhtin’s insistence on dialogue, we show that ethics should be conceptualized as a modality of interpersonal action rather than as a component of society or of the self (Lambek,

2010). Yet when it comes to decisions about right and wrong, we all want some kind of certainty.

The proliferation of principles, guidelines, standards, and checklists for ethical, responsible, or trustworthy AI is indicative of the search for certainty amid the doubts about the ability to ‘get it right’ in the design of AI systems. We do not argue against checklists or clear guidelines – these are important and necessary efforts. Rather, an *unfinalizable* approach to ethics acknowledges that the effort towards AI ethics does not simply *end* with a definitive checklist or set of principles but must continually embrace new doubts as part of the challenging task to cultivate certainty. Much like the ritual infrastructures, nodes of certainty can sustain these efforts through iteration and by providing structure for action and dialogue. Spaces for discussion and expression of concern enable mechanisms that allow nodes of certainty to be established as well as re-evaluated and questioned when needed. These mechanisms are iterative and participative, requiring collective action. Rather than aiming towards explicit solutions for ethical dilemmas as certainty, we suggest an *answerably* attainable path through spaces for doubt towards nodes of certainty.

It may not be possible to create a definitive and internationally recognized ethical framework for the design, production, use and governance of AI, as the European Group on Ethics in Science and New Technologies (EGE, 2018) has demanded, but attempts to do so are the very practice of ethics that is needed to make sense of AI and its possible futures.

Notes

1. While there are many definitions of AI, here we follow the terminology put forth in the IEEE Ethically Aligned Design report as a pragmatic solution as this was the point of departure for the discussions on which this paper is based.
2. The term ‘hackathon’ here refers to the specific name of the event “AI Hackathon: Ethical Dilemmas in AI” and our efforts to “hack ethics in AI” in the design of the event.
3. Although the term *nezavershennost* is typically translated from Russian as *unfinalizability*, it can also be translated as *unfinishness*, which is perhaps a better translation for our purposes implying that the work of ethics can never be ‘finished’ as it were.
4. The hackathon goals and structure are explained in detail in the hackathon report ‘Addressing Ethical Dilemmas in AI: Listening to Engineers’ created by the authors with the participation of partner organizations and accessible at <https://nordicengineers.org/2021/01/listen-to-the-engineers/>

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References

- AI HLEG (High-Level Expert Group on Artificial Intelligence). (2019). *A definition of Artificial Intelligence: Main capabilities and scientific disciplines*. European Commission.
- Association of Nordic Engineers [ANE]. (2018). *Nordic engineers' stand on Artificial Intelligence and ethics report*. <https://ipaper.ipapercms.dk/IDA/ane/report/>
- Association of Nordic Engineers [ANE]. (2021). *Addressing ethical dilemmas in AI: Listening to engineers*. <https://nordicengineers.org/2021/01/listen-to-the-engineers/>
- Bakhtin, M. (1993). *Toward a Philosophy of the Act*. (V. Liapunov, Ed., & M. Holquist, Trans.), Univ. Of Texas Press.
- Bakhtin, M. (ca 1920). *Автор и герой в эстетической деятельности*. [The Author and Hero in Aesthetic Activity]. <https://dustyattic.ru/articles/other-articles/author-and-hero/author-and-hero-problem-of-relations>
- Beaudouin, V., Bloch, I., Bounie, D., Cléménçon, S., d'Alché-Buc, F., Eagan, J., Maxwell, W., Mozharovskiy, P., & Parekh, J. (2020). Flexible and context-specific AI explainability: A multi-disciplinary approach. <http://arxiv.org/abs/2003.07703>.
- Collingridge, D. (1981). *The social control of technology*. Open University Press.
- European Group on Ethics in Science and New Technologies [EGE]. (2018). *Statement on Artificial Intelligence, robotics and 'autonomous' systems*. European Commission.
- Frauenberger, C., Rauhala, M., & Fitzpatrick, G. (2017). In-action ethics. *Interacting with Computers*, 29(2), 220–236. <https://doi.org/10.1093/iwc/iww024>.
- Green, B., & Viljoen, S. (2020, January 27–30). *Algorithmic realism: Expanding the boundaries of algorithmic thought*. Proceedings of the 2020 ACM Conference on Fairness, Accountability and Transparency (FAccT 2020) (pp. 19–31). Association for Computing Machinery, Barcelona, Spain. <https://doi.org/10.1145/3351095.3372840>
- Greene, D., Hoffmann, A. L., & Stark, L. (2019, January 8–11). *Better, Nicer, Clearer, Fairer: A Critical Assessment of the Movement for Ethical Artificial Intelligence and Machine Learning*. Proceedings of the 52nd Hawaii International Conference on System Sciences (HICSS). <https://doi.org/10.24251/HICSS.2019.258>
- Groves, C. (2015). Logic of choice or logic of care? Uncertainty, technological mediation and responsible innovation. *Nanoethics*, Dordrecht 9(3), 321–333. <http://dx.doi.org.ep.fjernaadgang.kb.dk/10.1007/s11569-015-0238-x> <https://doi.org/10.1007/s11569-015-0238-x>
- Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1(9), 389–399. <https://doi.org/10.1038/s42256-019-0088-2>
- Kroll, J. A., Huey, J., Barocas, S., Felten, E. W., Reidenberg, J. R., Robinson, D. G., & Yu, H. (2016). *Accountable algorithms* (SSRN Scholarly Paper ID 2765268). Social Science Research Network. <https://papers.ssrn.com/abstract=2765268>
- Lambek, M. J. (2010). *Ordinary ethics: Anthropology, language, and action*. Fordham University Press.
- Lampland, M., & Star, S. L. (Eds.). (2009). *Standards and their stories: How quantifying, classifying, and formalizing practices shape everyday life*. Cornell University Press.
- Madaio, M. A., Stark, L., Wortman Vaughan, J., & Wallach, H. (2020, April 25–30). *Co-Designing checklists to understand organizational challenges and opportunities around Fairness in AI*.

- Proceedings of the 2020 ACM Conference on Human Factors in Computing systems (pp. 1–14). Honolulu, Hawaii. <https://doi.org/10.1145/3313831.3376445>
- Mahieu, R., van Eck, N. J., van Putten, D., & van den Hoven, J. (2018). From dignity to security protocols: A scientometric analysis of digital ethics. *Ethics and Information Technology*, 20(3), 175–187. <https://doi.org/10.1007/s10676-018-9457-5>
- Metcalf, J., Moss, E., & Boyd, D. (2019). Owing ethics: Corporate logics, silicon valley, and the institutionalization of ethics. *Social Research: An International Quarterly*, 86(2), 449–476. <https://muse.jhu.edu/article/732185>
- Mittelstadt, B. (2019). Principles alone cannot guarantee ethical AI. *Nature Machine Intelligence*, 1(11), 501–507. <https://doi.org/10.1038/s42256-019-0114-4>
- Mittelstadt, B. D., Allo, P., Taddeo, M., Wachter, S., & Floridi, L. (2016). The ethics of algorithms: Mapping the debate. *Big Data & Society*, 3(2), 1–21. <https://doi.org/10.1177/2053951716679679>
- Morson, G. S., & Emerson, C. (1990). *Mikhail bakhtin: Creation of a prosaics*. Stanford University Press.
- Patil, D. J., Mason, H., & Loukides, M. (2018, July 17). *Of oaths and checklists*. O'Reilly Media. <https://www.oreilly.com/radar/of-oaths-and-checklists/>
- Pillai, A. G., Baki Kocaballi, A., Wah Leong, T., Calvo, R. A., Parvin, N., Shilton, K., Waycott, J., Fiesler, C., Havens, J. C., & Ahmadpour, N. (2021, May 8–13). *Co-designing resources for ethics education in HCI*. 2021 ACM conference on Human Factors in Computing systems. ACM, New York, NY, USA, Yokohama, Japan.
- Pink, S. (2019). Ethics in an uncertain world: Between theory and practice. In S. Pink, V. Fors, & T. O'Dell (Eds.), *Theoretical scholarship and applied practice* (pp. 29–50). Berghahn Books. <https://doi.org/10.2307/j.ctvw04bkj.6>
- Schiff, D., Biddle, J., Borenstein, J., & Laas, K. (2020, February 7–8). *What's next for AI ethics, policy, and governance? A global overview*. Proceedings of the AAAI/ACM Conference on AI, ethics, and society (pp. 153–158). New York, NY. <https://doi.org/10.1145/3375627.3375804>
- Sjørølev, I. (2017). Ritual infrastructure. *The Cambridge Journal of Anthropology*, 35(2), 65–78. <https://doi.org/10.3167/cja.2017.350206>
- Ustek-Spilda, F., Powell, A., & Nemorin, S. (2019). Engaging with ethics in internet of things: Imaginaries in the social milieu of technology developers. *Big Data & Society*, 6(2), 1–12. <https://doi.org/10.1177/2053951719879468>
- Vallor, S. (2011). Carebots and Caregivers: Sustaining the ethical ideal of care in the twenty-first century. *Philosophy & Technology*, 24(3), 251–268. <https://doi.org/10.1007/s13347-011-0015-x>
- Weeden, K. A. (2002). Why do some occupations pay more than others? Social closure and earnings inequality in the United States. *American Journal of Sociology*, 108(1), 55–101. <https://doi.org/10.1086/344121>
- Whittlestone, J., Nyrop, R., Alexandrova, A., Dihal, K., & Cave, S. (2019). *Ethical and societal implications of algorithms, data, and artificial intelligence: A roadmap for research*. Nuffield Foundation. <https://www.nuffieldfoundation.org/sites/default/files/files/Ethical-and-Societal-Implications-of-Data-and-AI-report-Nuffield-Foundation.pdf>
- Wieringa, M. (2020, January 27–30). *What to account for when accounting for algorithms: A systematic literature review on algorithmic accountability*. Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAccT 2020) (pp. 1–18). Barcelona, Spain. <https://doi.org/10.1145/3351095.3372833>